

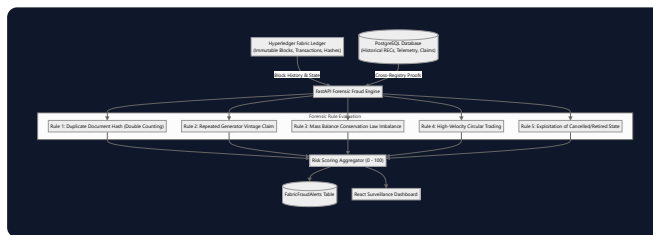
Forensic Fraud Detection Engine Specification

This document details the multi-engine forensic fraud detection architecture, cross-referencing on-chain distributed ledger states, transaction histories, and off-chain telemetry records.

1. Why AI/ML & Complex Fraud Models Stay Off-Chain

Smart contracts on Hyperledger Fabric must execute **strictly deterministically** across all endorsing peers. Every peer that executes a transaction proposal must arrive at the exact same read-write set down to the bit.

- Non-deterministic elements (floating-point neural network inference, asynchronous external API queries, time-of-day variations) cause **endorsement policy mismatch** and transaction rejection.
- Heavy computational tasks (deep learning, large graph cycle traversal) cause latency bottlenecks in consensus.
- Solution:** The Fraud Detection Engine operates in the **FastAPI layer**, consuming immutable transaction blocks from Fabric, cross-referencing off-chain databases, and recording high-severity alerts.



2. Core Forensic Rules & Heuristics

2.1 Rule 1: Duplicate Document Hash (Double Counting)

- Threat:** A rogue generator uploads the exact same generation proof PDF to mint multiple distinct certificates under different IDs or dates.
- Detection:** Queries `fabric_rec_records` for any existing REC with the same `document_hash` where `rec_id` $\neq$ `current_rec_id`.
- Impact:** Adds **+85.0 points** to risk score $\rightarrow$ triggers **CRITICAL** severity alert `DUPLICATE_DOCUMENT_HASH`.

2.2 Rule 2: Repeated Generation Period Claim

- Threat:** A solar farm claims 100 MWh for September 2026, and later submits a second claim for the exact same facility on the same generation date.
- Detection:** Checks if `(generator_id, generation_date)` has already been committed to the registry.
- Impact:** Adds **+40.0 points** to risk score $\rightarrow$ triggers **HIGH** severity alert `DUPLICATE_GENERATION_CLAIM`.

2.3 Rule 3: Mass Balance Conservation Law Imbalance

- **Threat:** Synthetically inflating certificate units during transfers or balance allocations.
- **Detection:** Validates:

$$\text{activeQuantity} + \text{retiredQuantity} \equiv \text{issuedQuantity}$$
$$\sum_{\text{participants}} \text{balances}[\text{participant}] \equiv \text{activeQuantity}$$

- **Impact:** Adds **+35.0 points** to risk score → triggers **HIGH** severity alert **QUANTITY_INCONSISTENCY**.

2.4 Rule 4: High Transaction Velocity & Wash Trading

- **Threat:** Artificial trading volume generated by rapidly transferring RECs back and forth between colluding entities.
- **Detection:** Analyzes the chronological transaction block history from **getRECHistory**. If more than 5 transfers occur in rapid succession or exhibit cyclical owner patterns (**A → B → A**), an alert is flagged.
- **Impact:** Adds **+15.0 points** to risk score.

2.5 Rule 5: Exploitation of Cancelled / Voided RECs

- **Threat:** Attempting to transfer, market, or retire certificates that have been cancelled by regulatory directive.
- **Detection:** Checks status on ledger. If **CANCELLED**, flag immediately.
- **Impact:** Adds **+20.0 points** to risk score.

3. Risk Level Scoring Framework

The composite risk score $S \in [0.0, 100.0]$ maps to regulatory triage tiers:

Score Range	Risk Level	Action Directive
0.0 – 34.9	LOW	Automatically approved for trading and settlement.
35.0 – 69.9	MEDIUM	Flagged for manual review by issuing body before retirement.
70.0 – 100.0	HIGH / CRITICAL	Suspended; automated alert dispatched to RegulatorOrg for investigation.